

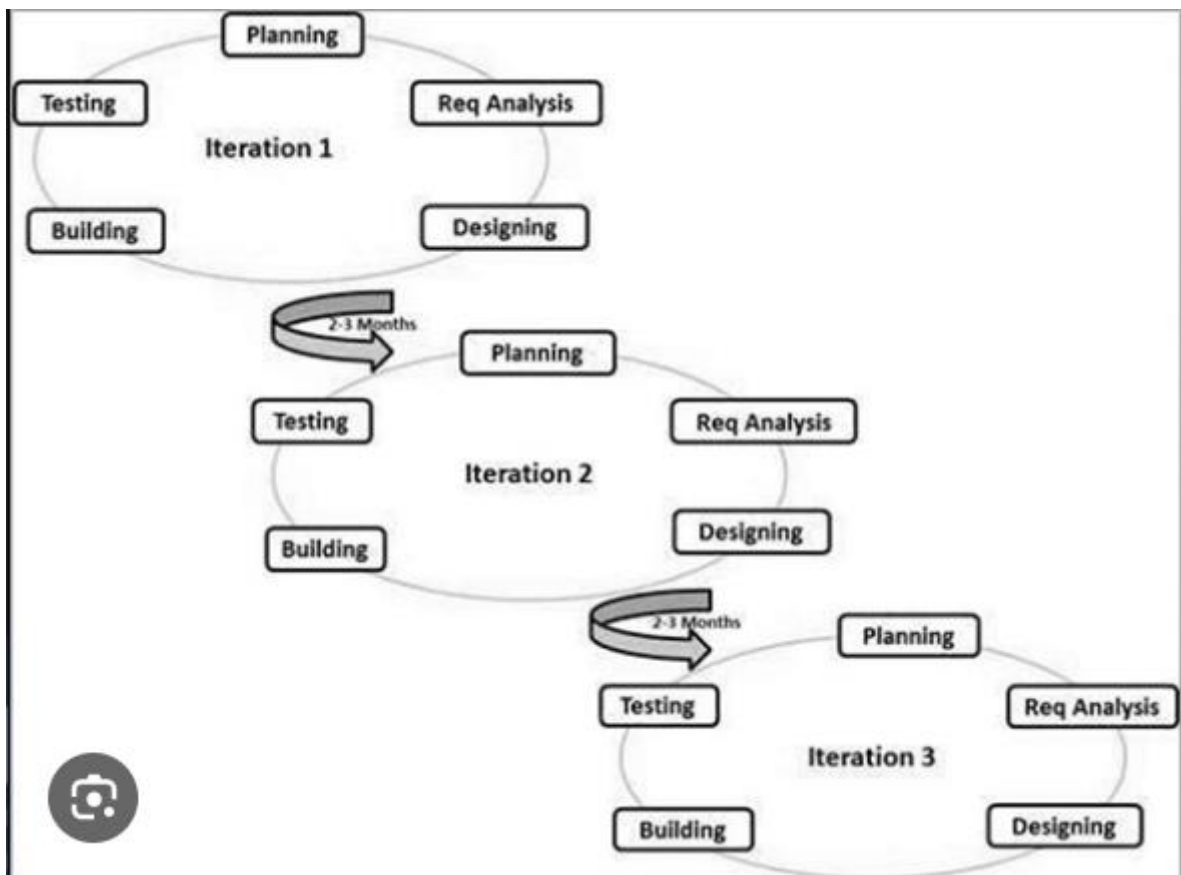
CHAPTER 7: Agile & Scrum Methodology

1. Introduction to Agile

What is Agile?

Agile is a modern project management and software development methodology that focuses on Flexibility, Collaboration, Fast delivery, Continuous improvement, Customer satisfaction

Instead of building a complete system at once, Agile divides the project into small parts delivered step by step.



Traditional Method vs Agile

Traditional Method (Waterfall)	Agile Method
Work done once from beginning to end	Work divided into small iterations
Changes are difficult	Changes are welcomed

Customer sees result at the end	Customer sees progress regularly
Long delivery time	Fast delivery
Heavy documentation	More collaboration

2. Agile Principles

Agile is based on 12 principles from the Agile Manifesto.

Simplified Agile Principles

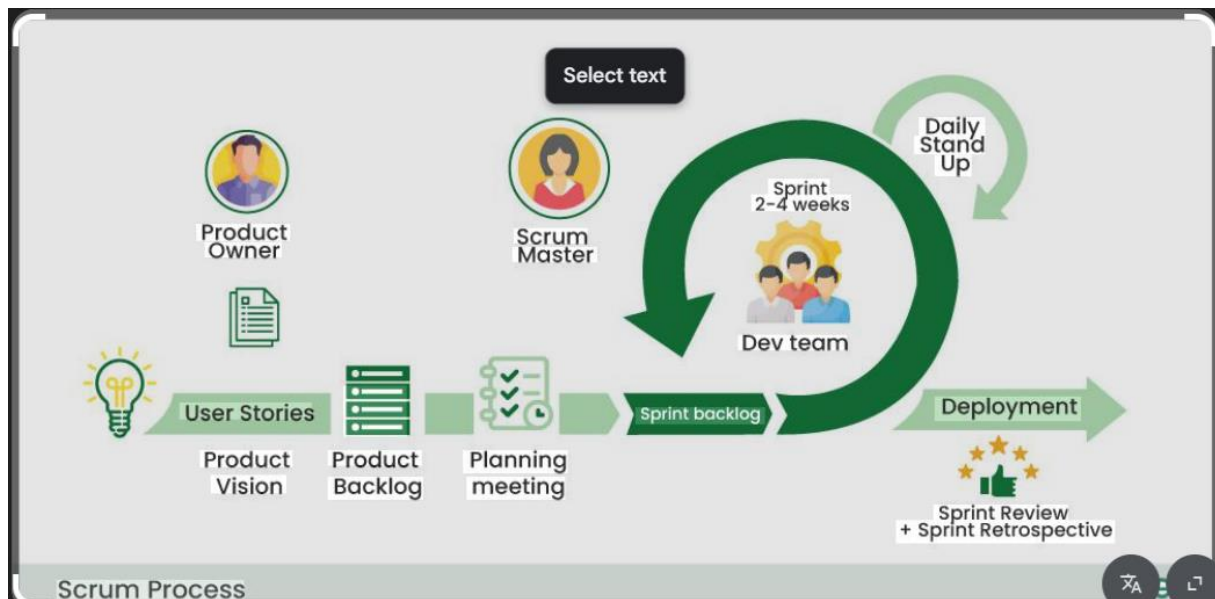
- 1. Customer Satisfaction:** Deliver useful software regularly.
- 2. Welcome Changes:** Even late changes are accepted if they improve the product.
- 3. Frequent Delivery:** Deliver small working parts quickly.
- 4. Collaboration:** Developers and business people work together.
- 5. Motivated Teams:** Trust and support team members.
- 6. Face-to-Face Communication:** Direct communication is preferred.
- 7. Working Product First:** A working application is more important than too much documentation.
- 8. Sustainable Development:** Work at a constant and healthy pace.
- 9. Technical Excellence:** Good design and quality matter.
- 10. Simplicity:** Avoid unnecessary work.
- 11. Self-Organizing Teams:** Teams decide how to accomplish work.
- 12. Continuous Improvement:** Teams regularly evaluate and improve themselves.

3. What is Scrum?

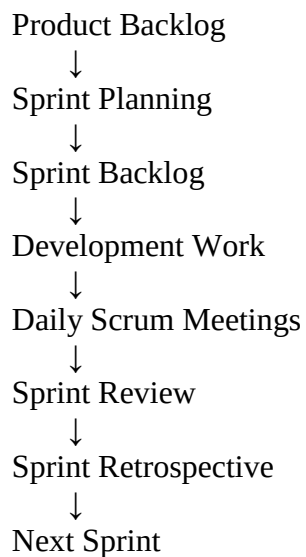
Definition--Scrum is the most popular Agile framework used to manage projects. It organizes work into short periods called **Sprints**.

A Sprint usually lasts: 1 week, 2 weeks or 4 weeks.

At the end of each Sprint, a small working product is delivered.



4. Scrum Workflow



5. Scrum Roles

A. Product Owner

Responsibilities: Represents the customer, Defines product requirements, Prioritizes tasks, Manages Product Backlog

Example--In an e-commerce application: Decides that payment system is more important than chat feature.

B. Scrum Master

Responsibilities: Facilitates Scrum process, Removes obstacles, Ensures Agile principles are respected, Helps team improve.

Example--If developers lack tools or permissions, Scrum Master solves the issue.

C. Development Team

Responsibilities: Designs the system, Codes the application, Tests the product, Delivers working features.

Characteristics--Self-organized, Collaborative, Multi-skilled.

6. Important Scrum Concepts

A. Product Backlog

A prioritized list of all project requirements.

Example

ID	Requirement	Priority
1	User login	High
2	Shopping cart	High
3	Dark mode	Low

B. Sprint Backlog

Tasks selected from Product Backlog for the current Sprint.

C. Sprint

A short development cycle.

Example

Sprint Duration : 2 weeks

Goal: Build user authentication system

D. Daily Scrum

A short daily meeting (15 minutes).

Each member answers:

1. What did I do yesterday?
2. What will I do today?
3. What problems am I facing?

E. Sprint Review

Team demonstrates completed work.

F. Sprint Retrospective

Team discusses: What went well, What failed, What can improve.

7. Sprint Planning

Definition--Sprint Planning is a meeting where the team selects tasks to complete during the Sprint.

Steps in Sprint Planning

Step 1: Review Product Backlog

Select important features.

Step 2: Define Sprint Goal

Example: "Create a secure login system."

Step 3: Break Work into Tasks

Example:

- ✚ Design login page
- ✚ Create database table
- ✚ Develop authentication API
- ✚ Test login security

Step 4: Estimate Time

Example:

- ✚ Login UI → 4 hours
- ✚ API → 6 hours

PRACTICE

ACTIVITY 1: Create Product Backlog

A company wants a **Student Management Web Application**.

The system should Register students, Manage courses, Store grades, Generate reports, Allow login.

Step 1: Identify Features

Possible features: User authentication, Student registration, Course management, Grade management, PDF report generation.

Step 2: Prioritize Features

Example Product Backlog

Backlog ID	User Story	Priority	Estimated Effort
PB1	User can login	High	5
PB2	Admin can add students	High	8
PB3	Admin can add courses	Medium	6
PB4	Teacher can enter grades	High	7
PB5	Generate PDF reports	Medium	9
PB6	Dark mode	Low	3

ACTIVITY 2: Define User Stories

What is a User Story?

A short description of a feature from the user's perspective.

User Story Format

As a [user],
I want [feature],
So that [benefit].

Examples

Example 1

As a student,
I want to login securely,
So that my data is protected.

Example 2

As an administrator,
I want to register students,
So that student records are stored digitally.

Example 3

As a teacher,
I want to upload grades,
So that students can see their results.

ACTIVITY 3: Simulate Sprint Planning

Sprint Scenario

Sprint Duration: 2 Weeks

Team Capacity: 30 Hours

Step 1: Select Backlog Items

Selected:

-  Login system
-  Student registration

Step 2: Define Sprint Goal

Build secure login and student registration module.

Step 3: Create Sprint Tasks

Total Hours = 27 Hours

Task	Assigned To	Hours
Design login page	Alice	4
Create login API	Bob	6
Create database table	John	3
Build registration form	Alice	5
Connect form to database	Bob	5
Testing	Team	4

Step 4: Daily Scrum Simulation

Day 1

Alice

- 🚦 Completed UI design
- 🚦 Will start registration form
- 🚦 No blockers

Bob

- 🚦 Started login API
- 🚦 Needs database credentials

John

- 🚦 Created database schema
- 🚦 No blockers

DELIVERABLE

1. Product Backlog

ID	Feature	Priority
PB1	Login System	High

PB2	Student Registration	High
PB3	Course Management	Medium
PB4	Grade Upload	High
PB5	PDF Reports	Medium

2. Sprint Plan

Sprint Goal

Develop authentication and registration modules.

Sprint Tasks

Task	Responsible	Duration
Login UI	Alice	4h
Login Backend	Bob	6h
Database Setup	John	3h
Registration Form	Alice	5h
Database Connection	Bob	5h
Testing	Team	4h

CASE STUDY

A startup wants to build an online food delivery application.

Features: Customer registration, Restaurant listing, Online ordering, Payment integration , Order tracking

Questions

1. Define Agile methodology.
2. Explain Scrum.
3. Name three Scrum roles.
4. What is a Sprint?
5. What is a Product Backlog?
6. Write two user stories.
7. Define Sprint Planning.
8. What is Daily Scrum?
9. Why is Agile flexible?
10. Create a mini Product Backlog for the application.

MCQs

1. Scrum is:

A. Database B. Agile framework C. Programming language D. Operating system
2. A Sprint is:

A. Meeting B. Document C. Short development cycle D. Database
3. The Product Owner:

A. Tests hardware B. Cleans database
C. Prioritizes requirements D. Installs software
4. Daily Scrum usually lasts:

A. 2 hours B. 1 day C. 15 minutes D. 5 hours
5. User stories describe:

A. Server configuration B. User needs C. Network cables D. Antivirus setup